

A105 – The Ladder Base Unit N_0

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.1 What N_0 Is

[B] A100 showed that the rough ξ -ladder of the lepton masses carries a small, generation-dependent correction. Writing the correction required per rung as an effective factor N_g via $K = 1 - N_g \xi$, the three factors stand in the ratio

$$N_1 : N_2 : N_3 = 1 : 1.981 : 2.971 \approx 1 : 2 : 3.$$

The correction thus grows *linearly with generation number*,

$$\boxed{N_g \approx g \cdot N_0}, \quad N_0 \approx 38.5,$$

and the three individual estimates $N_1, N_2/2, N_3/3$ lie at 38.89, 38.53, 38.52 – consistent with each other to under one per cent. That is the core: N_0 is *not a fixed constant of the theory* but the *ladder step per generation*. At $g = 2$ it is already 77, at $g = 3$ about 116.

[K] This triple consistency is, however, weaker than it looks: only *two* of the three estimates are independent. Because a consistently multiplicative correction $K^{p_3} = K^{p_1} K^{p_2}$ and hence $p_3 = p_1 + p_2$ forces $N_3 \approx N_1 + N_2$ arithmetically (checked: 115.95 vs. 115.55, difference 0.35 %, second order in ξ) – in the ratio 1:1.981:2.971 only 1.981 is measured, the third number is fixed as $1 + 1.981$. The generation-linear law therefore rests on *one* observation (the μ/e rung), not three; it is a candidate on one measurement, not a triply supported regularity. The standard from A220 – “do not count dependent as independent” – applies immediately here.

[K] Two things follow immediately. First, N_0 is to be sharply separated from the fixed number 100 in $K_{\text{frak}} = 1 - 100\xi$ (A040): the 100 is the *one* RG scale of a corpus-wide constant, N_0 is a generation counter in a different sector. That both lie near powers of ten does not make them the same quantity. Second, with the generation-linear law the real question is not the *value* of N_0 but why the ladder is corrected in a *generation-counting* way at all. A closed form for the base alone would leave this law untouched.

.2 The Right Standard: Determination Tolerance Instead of Last Digit

[K] N_0 is not known to arbitrary precision. In the referenced mode its value follows solely from the two τ rungs, and their precision is limited by the τ mass (± 0.12 MeV). Propagating that gives

$$N_0 = 38.52 \pm 0.21,$$

a relative determination uncertainty of about half a per cent. From this follows the standard against which every candidate expression is to be measured: not whether it hits 38.5 to the last decimal – that is in principle unreachable – but whether it *falls within this interval*.

[K] This standard has a converse that carries the whole section: *the band is wide enough that many expressions fall into it*. Hitting a ± 0.21 window is easy. Hence bare compatibility with the band *cannot* be the test that turns a candidate into a derivation. Compatibility says the number fits; it does not say the number *follows*.

.3 The Candidates

[K] Three expressions lie in or near the band and are to be recorded as *permissible, unrefuted* candidates – none is a derivation:

Expression	Value	Distance
$100/\varphi^2$	38.20	-1.5σ
$3 \cdot (\pi^2/2) \cdot \varphi^2$	38.76	$+1.1\sigma$
$\ln(M_{\text{Pl}}/\nu)$	38.44	-0.4σ
$100/e$	36.79	-8.3σ

with $\varphi = \frac{1+\sqrt{5}}{2}$ and $V_{S^4} = \pi^2/2$ the volume of the unit 4-ball. The first three are compatible; $100/e$ lies far outside and is excluded. The middle expression $3 \cdot (\pi^2/2) \cdot \varphi^2$ lies even closer to the centre than $100/\varphi^2$.

[K] And here the standard from the previous section applies: that three seemingly well-motivated expressions fall into the band is precisely the diagnosis. A fourth would do the same. Band hits do not separate candidates because the band is wide.

.4 Why None of Them Is a Derivation

[K] The objection falls equally on all three expressions, and it concerns not the *value* but the *form*. For none is there a justification of why precisely this form should be the form of N_0 : why the 4-ball volume $\pi^2/2$, why φ^2 , why the factor 3 and not 9. What is missing is a chain that begins at the T^4/Z_3 geometry and has this structure *as output at the end* – rather than an expression assembled backwards to hit an already known value.

[K] The role of φ is to be precisely stated, because φ is not foreign to the mass sector. In A120 φ appears *forward and parameter-free* as an eigenvalue of a named construction – the golden inflation $\varphi^2 = \varphi + 1$ in the icosahedral redistribution. In *this* role φ is permissible and derived. Strictly to be separated from this is φ as a *power inserted into an expression to hit a scale* – φ^N as a fitting base. That is exactly the warning example: it carries no forward content. Both $100/\varphi^2$ and $3 \cdot (\pi^2/2) \cdot \varphi^2$ are of the second kind – φ^2 assembled, not produced by a construction. That is why both remain candidates and do not become derivations, even though φ itself is at home in the theory when it appears as output of a named symmetry.

.5 A Warning Example: Target First, Justification After

[K] How easily a fit dresses itself as a derivation is shown by a concrete case. To the expression $3 \cdot (\pi^2/2) \cdot \varphi^2 = 38.76$, which lies $+1.1\sigma$ above the centre, an “anomaly” term Δ was subsequently added to close the residual. Its value:

$$\Delta = \frac{3}{4\pi} = 0.2387,$$

and the gap to be closed is $38.758 - 38.52 = 0.238$ – the same number to three figures. The correction was therefore not derived and then found to fit; the gap was known first, and a term of the right size was *subsequently* given a name. Two physical features confirm this independently of the coincidence: a one-loop contribution would carry the factor $1/(4\pi)^2$, not linear $1/(4\pi)$; and a genuine localised anomaly would be *quantised* – an integer index, not a fractional 0.2387. The physical dress does not fit the number underneath it. That does not refute 38.52 as a candidate; it only shows that the anomaly narrative is not the reason why the value sits there.

[K] From this follows the test rule that A220 records generally and that applies particularly here: a derivation is not to be assumed in prose but to be written as a script that begins at the geometry, contains *no target value*, and one checks what number it prints. Prose can hide a fitted step behind a secure sentence; code cannot. A value hitting the centre of a ± 0.21 band to the last digit is not a triumph but a warning: nothing fundamental lands exactly on the mean of a measurement unless it was built to do so.

.6 Verification Script

- The tolerance $N_0 = 38.52 \pm 0.21$, the σ distances of the four candidates, the target back-computation $\Delta = 3/(4\pi)$ vs. the gap, and the generation run $N_g = gN_0$:
python/A_Serie_Skripte/a105_n0.py

.7 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
N_0	Recursion limit; value at which the iteration converges
δ	Refraction strength; deviation from ideal torus lattice
Δ_{rec}	Recursion deficit; difference to next closure

.8 Open Questions

First, N_0 is open. It is well bounded as a value within its tolerance, but not derived from the geometry. No expression presented so far is more than a permissible candidate.

Second, the standard is named. The question is closed by a construction that begins at the T^4/Z_3 geometry, fixes invariant, exponent, and scales, and delivers N_0 at the end – a construction that could have missed and did not –, not by another well-looking expression that falls into the band.

Third, the generation law remains the deeper task. Even a perfect closed form for the base does not explain why the correction $N_g = g N_0$ runs generation-counting. That is the real open question, and it lies beneath the question of the numerical value.

.9 Supersedes

This document subsumes: Dok. 292 (generation-linear coupling, $N_g = g N_0$, tolerance $N_0 = 38.52 \pm 0.21$, candidates $100/\varphi^2$, $100/e$), Dok. 190 (R54/R55: N_0 open, separation $K_{\text{frak}}-100 / N_0$; P35). Incorporated: the external candidate $3 \cdot (\pi^2/2) \cdot \varphi^2$ and the $\Delta = 3/(4\pi)$ target back-computation from the IPI correspondence (July 2026), booked as permissible-but-not-derived candidate and as warning example respectively.

.10 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.